

# Gonçalo L. Fernandes

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## SUMMARY

AI/ML Developer building **production-grade intelligent systems** on Azure Cloud. I design, train, and deploy AI pipelines that process real enterprise data, specializing in **multi-model financial document processing** with **OCR, NLP**, and **GenAI**, as well as in custom **Computer Vision** and **Deep Learning**, for B2B enterprise clients across European markets. In Computer Vision, I build and train custom **object detection pipelines** for drone-based warehouse inventory auditing, fine-tuned end-to-end on real client data.

## SKILLS

**AI/ML & Document Processing:** Azure Document Intelligence, Azure AI Foundry, scikit-learn, Custom AI Models, ML Model Training, Document Classification, OCR, IDP, Multi-Model AI, AI Orchestration, Pattern Recognition

**Computer Vision & Deep Learning:** PyTorch, TensorFlow, Ultralytics YOLO, OpenCV, ONNX, LayoutLMv3, Object Detection, Image Classification, Fine-tuning, Custom Training Pipelines, DeepSORT

**LLM & GenAI:** GPT-4.1, Anthropic Claude API, Meta Llama, Hugging Face, Prompt Engineering, LLM Integration, LLM Cost Optimization, Local LLM Deployment, Semantic Analysis, NLP

**Cloud & DevOps:** Azure Cloud, Azure Logic Apps, Azure Blob Storage, Azure DevOps, Azure API Management, Azure Static Web Apps, Docker, CI/CD Pipelines, Cloud Architecture

**Development:** Python, FastAPI, Angular 19, TypeScript, NumPy, Pandas, Flutter, PostgreSQL, REST APIs, OpenAPI, Microservices, API Integration, JSON, Real-time Systems

**Tools & Platforms:** Microsoft Graph API, Azure OpenAI Service, Azure Machine Learning, Git, Redis, Power BI

## WORK EXPERIENCE

### COSMOS-OMNI

Mar 2025 – Present

AI/ML Developer — Permanent (Aug 2026 – Present)

Porto Metropolitan Area · Hybrid

Permanent position after one year as Junior Developer; continued ownership of the **AEI (Accounting Engine Interface) document processing ecosystem** (Azure Logic Apps, Azure Document Intelligence, Azure AI Foundry, GPT-4.1), the **Angular 19 enterprise document portal**, and the drone-based **Computer Vision warehouse auditing pipeline** for fashion retail.

AI/ML Junior Developer (Aug 2025 – Jul 2026)

- Development and deployment of **AI automation systems** for document processing and invoice classification, with **20+ production cases** and **24 custom extraction models** in Azure Logic Apps; real-time processing with **65% reduction in manual errors** and **40% reduction in processing time**.
- **Azure Document Intelligence** integration, custom model training and classification model retraining in **Azure AI Foundry**; multi-model AI orchestration and LLM integration (**GPT-4.1**) for production environments.
- Frontend migration of an enterprise document management portal from legacy ASP.NET MVC to **Angular 19 SPA**, with **OAuth2 Authorization Code + PKCE** authentication through **Azure API Management**.
- **Computer Vision pipeline** for drone-based warehouse auditing: custom-trained **YOLO26** detection models (4 classes, ~12,000 annotations) validated end-to-end on real imagery from a fashion retail client; batch inference with **Azure Blob Storage** integration and WMS submission via Azure API Management.
- **Azure DevOps CI/CD** pipelines and **Azure Static Web Apps** deployment; Blob Storage pipeline management and standardized field taxonomies for API integration.

AI/ML Intern Developer (Mar 2025 – Jul 2025)

- API integration and automated deployment pipelines; **Azure Logic Apps** and **Blob Storage** workflow development; Document AI model training and optimization.
- Process automation and workflow orchestration; real-time systems development and monitoring; scalable architecture design and implementation.

Andreia Ferreira Serviços & Consultoria

Jul 2026 – Present

## Process Automation & IT Support (Freelance)

Vila Nova de Famalicão, Portugal · Hybrid

- Independent service provider for an administrative and solicitadoria practice supporting citizens and businesses with social security, tax, registry and documentation processes; **standardised workflow** for translating and normalising foreign tax documentation, reducing turnaround time on cross-border cases.
- Maintenance, repair and configuration of IT equipment: desktops, laptops, printers and peripherals.

## Worten Portugal

Feb 2025 – Aug 2025

### Technical Sales Representative in IT (Part-time)

Porto, Portugal

- Technical consultation and product recommendations based on customer needs at Portugal's leading tech retailer (+**230 stores**, +**4,000 employees**); stock and display management, knowledge updates on technology trends, promotion of additional services (Worten Resolve, warranties, insurance), and customer relationship building with loyalty programs.

## Ibersol Group

Aug 2024 – Feb 2025

### Customer Service in Hospitality (Part-time)

Vila Nova de Gaia, Portugal

- Promotion of services and products to enhance customer experience at Portugal's leading organized restaurant group; attention to detail in service delivery and presentation, customer loyalty initiatives, and teamwork in a fast-paced environment.

## Famous Determination

Jul 2023 – Oct 2023

### Electrical Collaborator

Portugal (nationwide)

- Electrical installations in residential and commercial buildings, supporting senior electricians in maintenance and repair work across Portugal; installation of lighting, outlets, and distribution panels with **RTIEBT compliance** (Portuguese electrical safety regulations).

## PC & Companhia

Mar 2023 – Jul 2023

### IT Technician (Internship)

Porto, Portugal

- OS installation and configuration (Windows, macOS, Linux); maintenance and repair of computers, laptops, and electronic equipment; hardware troubleshooting, diagnosis of hardware/software failures, and data backup/recovery services at a specialized computer repair company.

## PROJECTS

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All projects developed at COSMOS-OMNI.

### SkyVision Auditor — Drone-Based Warehouse Auditing with Computer Vision

Computer Vision | YOLO26 Custom | PyTorch | Ultralytics | Python | Azure Blob Storage | Azure APIM | Power BI

- End-to-end Computer Vision pipeline automating physical warehouse audits with drone-captured imagery across multi-zone fashion retail operations (~**6,000 SKUs**, ~**50,000 units**), transforming **multi-hour manual audits into a sub-1-hour automated scan** with CV processing kept on-premise.
- Custom **YOLO26** detection model (4 classes: box, clothing, footwear, empty\_space) with iterative dataset versioning; **filename-anchored location mapping** tying each frame to structured WMS codes — no barcodes, ArUco markers, or 3D scene reconstruction required.
- **6-level fill classification engine** via area-intersection geometry between detected stock and bin containers; resilient WMS submission through **Azure API Management** with local buffering and exponential retry; idempotent processing keyed by filename.

### Multi-Model Invoice Classification and Processing System

Azure Cloud | REST APIs | Machine Learning Classification | Azure OpenAI (GPT-4.1)

- **19 specialized AI models** (1 classifier + 9 custom extraction + 8 hybrid CU+LLM pipelines + 1 generic fallback) for an Iberian retail client, achieving **95%+ classification** and **97%+ extraction accuracy** across PT/ES markets.
- **Reduced manual processing by 90%** with scalable architecture delivering **20–25% higher accuracy** vs. generic solutions; supports unlimited new suppliers without redesign.

### Dual-Engine IDP Pipeline for Construction Industry

Azure Cloud | Azure OpenAI | REST APIs

- End-to-end IDP combining custom classification/extraction models with **Azure AI Foundry Content Understanding** and **GPT-4.1** semantic enrichment for a Portuguese construction company, **processing 30+ suppliers**.
- Parallel processing architecture with concurrent extraction branches converging into unified JSON output, plus per-invocation cost tracking.

Automated Financial Document Processing Pipeline

Azure Cloud | REST APIs

- End-to-end automation for invoices and delivery notes across multilingual documents, reducing processing time from **30 min to <2 min per document (95% reduction)** with **98%+ accuracy** at **200+ docs/day** with zero manual intervention.
- Fault-tolerant pipeline with automatic error handling, incidence tracking, and multi-tier archiving for compliance.

ML Document Extraction System

LayoutLMv3 | PaddleOCR | FastAPI | PyTorch | Docker

- Fine-tuned **LayoutLMv3** on supplier-specific datasets as a local alternative to cloud-based Azure AI services, targeting invoice extraction across PT/ES markets with full infrastructure control.
- Two-stage architecture (classification → category-specific extraction) with **PaddleOCR** as decoupled OCR backbone; containerised stack with Docker for reproducible deployment.

AI Warehouse Management System with Dual LLM Architecture

Spring Boot | Flutter | PostgreSQL | Docker | NLP — Scale: 6,100+ products, 40,000 items

- Full-stack warehouse system with **dual-AI architecture** intelligently switching local/cloud LLMs by time-of-day, achieving **79% AI cost reduction**.
- Custom **NL-to-SQL engine** with multilingual support (PT/ES/EN), voice-controlled hands-free operation (Speech-to-Text + Text-to-Speech), and real-time inventory visibility across 40,000+ items.

EDUCATION

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|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>ISTEC (Institute of Advanced Technologies PT)</b><br><i>Bachelor's Degree, Computer Engineering (EQF Level 6)</i>                                   | <b>Oct 2026 – Jul 2029</b> |
| <b>ISTEC (Institute of Advanced Technologies PT)</b><br><i>Associate's Degree, Networks and Computer Systems (EQF Level 5) — Grade: 16/20</i>          | <b>Oct 2023 – Jul 2026</b> |
| <b>IEFP — Instituto do Emprego e Formação Profissional</b><br><i>Certificate of Qualifications, Digital Commerce "Business Strategy" (EQF Level 4)</i> | <b>Feb 2024 – Jul 2024</b> |
| <b>IPTA — Instituto Profissional de Tecnologias Avançadas</b><br><i>IT Equipment Management (EQF Level 4) — Grade: 16/20</i>                           | <b>Sep 2020 – Jul 2023</b> |
| <b>Conservatório Regional de Música de Ferreirim</b><br><i>Euphonium Performance (5th Grade)</i>                                                       | <b>Sep 2014 – Jun 2020</b> |

ADDITIONAL

**Languages:** Portuguese (Native) | English (Upper Intermediate, B2) | Spanish (Basic, A2)  
**Driving License:** Category B (valid)  
**Availability:** Upon request